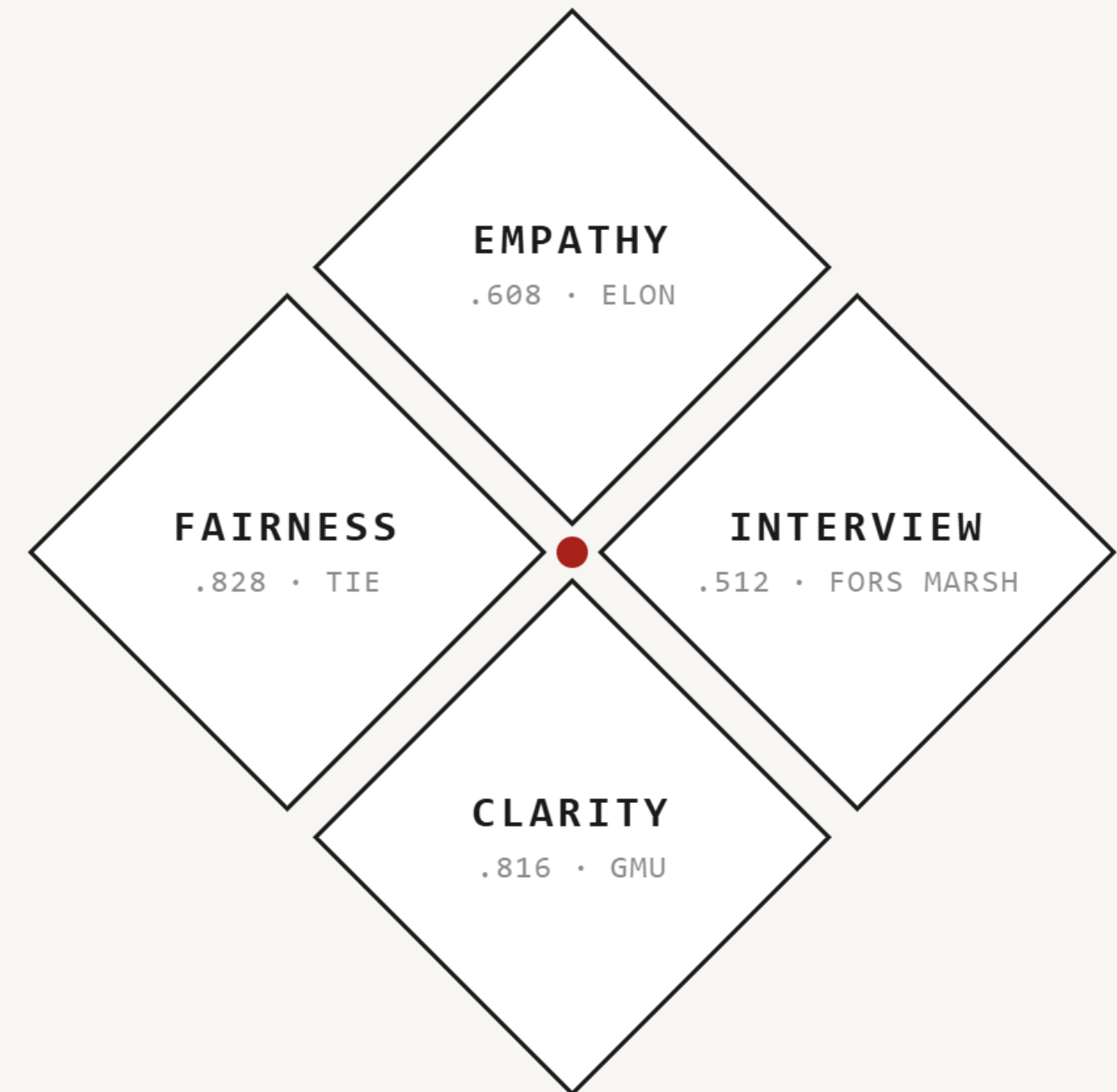


SIOP 2024 · MACHINE LEARNING COMPETITION

Retrospective

Four benchmarks, four metrics, four different winning teams.

What four winning approaches teach about applying LLMs to industrial-organizational problems.



.666

Winning composite PAID Team · George Mason University

4

Different metrics rolled into one composite

2/4

Tasks the overall winner did *not* win

THE COMPOSITE

$$\text{composite} = 0.25 \times (\text{empathy_acc} + \text{interview_cos} + \text{clarity_r} + \text{fairness_acc})$$

Four I-O benchmarks, one composite score

SIOP 2024 distributed four independent ML benchmarks tied to industrial-organizational organizational psychology tasks. Each had its own metric and its own data shape. shape.

The composite is a **0.25-weighted sum** — so a balanced submission beats a specialist.

Nobody won all four

Four benchmarks, four winning teams. The overall winner **lost on two of four tasks**.

THE PATTERN

PAID won without winning the most tasks — it took the two highest-ceiling tasks (Clarity, Fairness) and **absorbed a last-place finish on Interview**. A team that won *every* task would score just .691 — barely above PAID's .666.

	PAID Team George Mason	Akben & Aaron Elon	Hungry Llama Fors Marsh	Wonderlic Wonderlic Inc.	This repo PROJECTED
Empathy ACCURACY	.580	.608	.560	.488	~.58
Interview COSINE SIM	.440	.496	.512	.460	~.48
Clarity PEARSON R	.816	.676	.740	.772	~.70
Fairness ACCURACY	.828	.828	.760	.792	~.82
COMPOSITE	.666	.652	.643	.630	~.65
	◆ 1ST OVERALL	2nd	3rd	4th	est.

Highest score on each task among the four finalists. Fairness tied — tied — PAID & Akben, .828.

“This repo” = projected real-data estimate. Synthetic-run results are results are on slides 11–12 (not comparable).

BINARY CLASSIFICATION · METRIC: ACCURACY

Can a model tell empathy from coldness?

Binary classification of workplace feedback emails written to a colleague — Jonathan, on his work on his work on the Beta project. Each email is labelled empathetic or cold.

SAMPLE #041

LABEL: EMPATHETIC

“Jonathan — I know the Beta launch has been a grind. The dashboard work genuinely moved the needle and the team noticed. Take the next review off your plate; I’ve got it covered.”

SAMPLE #118

LABEL: COLD

“Jonathan, the Beta numbers are late again. Send the dashboard by EOD. We can’t keep adjusting the timeline around one person’s pace.”

WHY IT’S DIFFICULT

The score range was tight (.49–.61). The label distribution distribution itself is noisy — human raters disagreed on disagreed on borderline cases.

Akben & AAron



WON THIS TASK

Pairwise comparison via Elo rating

METRIC	accuracy
TRAIN SET	~200
BEST 2024	.608

Range across teams .49–.61



GENERATION · METRIC: COSINE SIMILARITY

Can a model sound like a specific candidate?

Given four prior question-and-response pairs from a candidate’s interview, generate the fifth response.

METRIC	cosine similarity
TRAIN SET	none
BEST 2024	.512

THE TWIST

The metric rewards **style match, not content quality**. The overall winner (PAID) finished *last* here by pooling responses across candidates.

ONE CANDIDATE · FIVE EXCHANGES

Q1	Tell me about a project you’re proud of.
R1	“I rebuilt our onboarding flow end-to-end — shipped it solo in three weeks.”
Q2	How do you handle disagreement on a team?
R2	“I push hard in the room, then commit fully once we decide.”
Q3–Q4	... two further exchanges in the same direct, high-ownership voice ...
Q5	Where do you want to be in five years?
R5	model generates this → must match R1–R4 voice

Hungry Llama

ForsMarsh

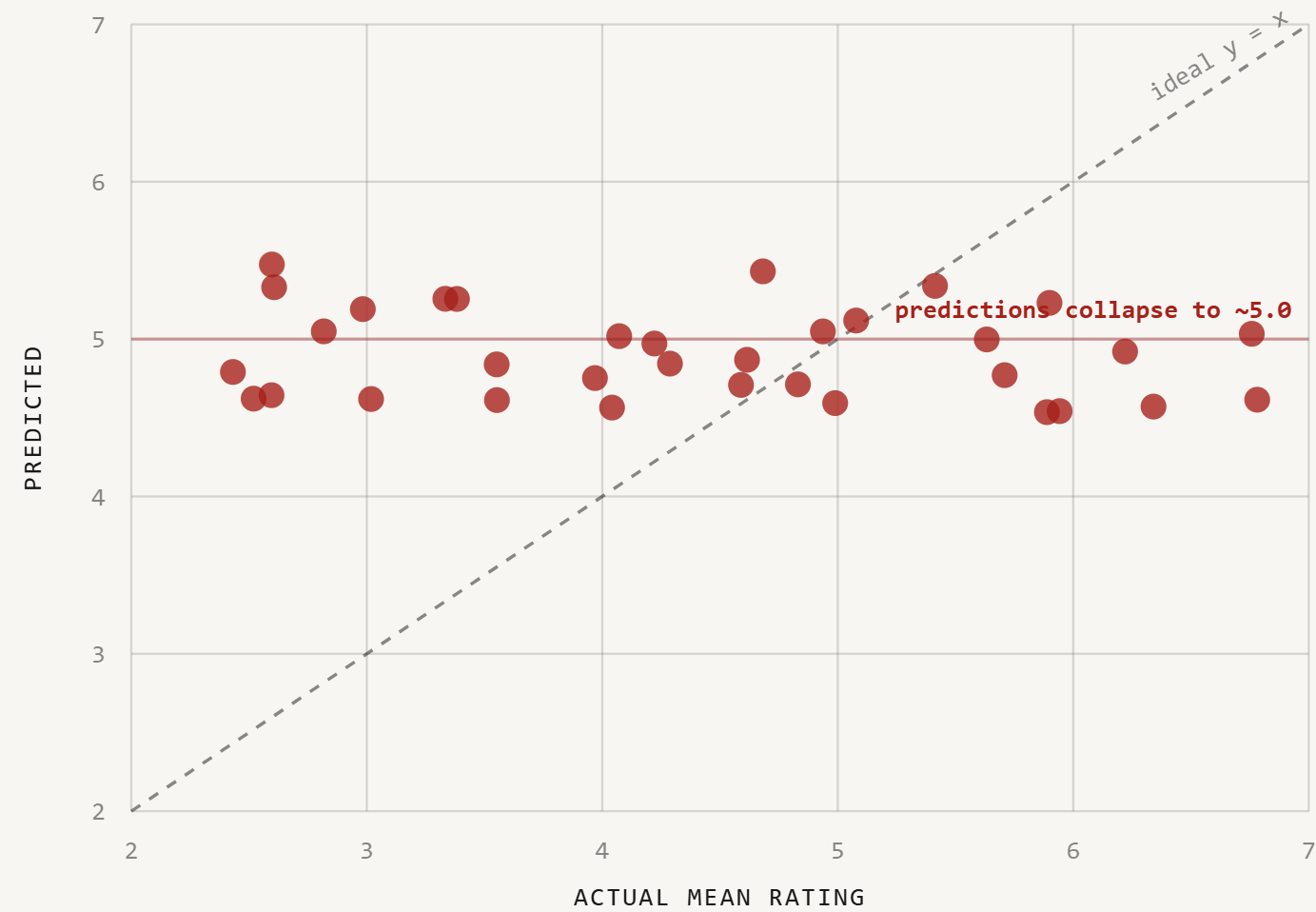
WON THIS TASK

Big-5 personality conditioning

REGRESSION · METRIC: PEARSON R

Can a model predict calibrated ratings? ratings?

Regression on personality-test items. Score on Pearson r between predicted and actual mean ratings.



METRIC	Pearson r
TRAIN SET	~200
BEST 2024	.816

THE UNDERSPREAD PROBLEM

LLMs cluster numeric predictions tightly around **5.0** — too narrow for Pearson r to land high. PAID abandoned GPT and fine-tuned **DeBERTa-v3-base** instead.

“Tried GPT4 (failed). Tried fine-tuned GPT3.5 (failed).”

PAID TEAM · COMPETITION DECK

PAID Team



WON THIS TASK

Fine-tuned regressor



PAIRED CLASSIFICATION · METRIC: ACCURACY

Which policy did raters think was fairer?

Given two policy descriptions, pick the one human raters voted fairer. The smallest task in the competition — only ~24 labelled pairs.

RATER MAJORITY ▾

POLICY · FIRST

Remote-first. Employees choose where they work; the company funds quarterly in-person summits for the whole team.

vs

POLICY · SECOND

Hybrid mandate. Three fixed in-office days each week; employees receive a \$500 annual home-office stipend.

TRAIN PAIRS

~24 smallest

BEST 2024

.828 tied

MOST-CONVERGED TASK

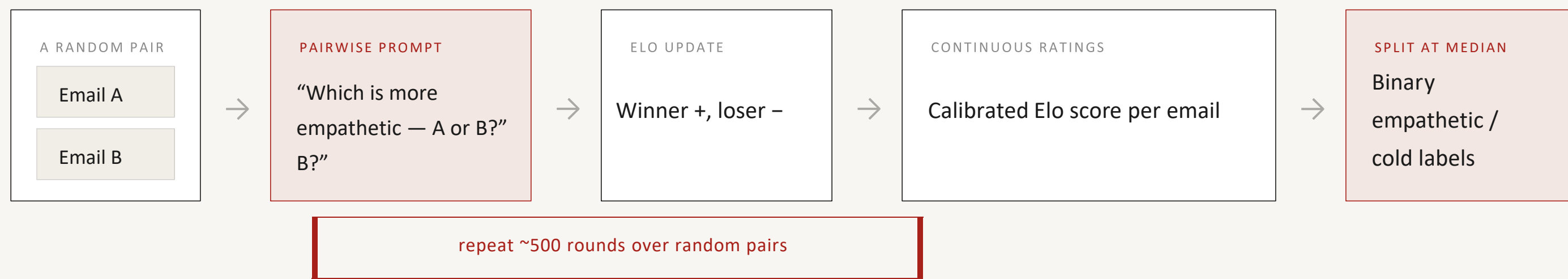
All four winners did roughly the same thing. Model choice mattered more than prompting — Mixtral hit .76 while GPT-4 hit .83.

Tied for first: PAID Team (GMU) & Akben & Aaron (Elon).

TRANSFERABLE MOVE · 01 / 3 · FROM AKBEN & AARON

Pairwise comparison

Use this when human labels are noisy and rater disagreement is high.



WHY IT WORKS

Asking “*is this empathetic?*” is harder than “*which of these two is more empathetic?*” Relative judgment is more reliable; Elo aggregation turns it into calibrated continuous ratings.

Akben & Aaron



TRANSFERABLE MOVE · 02 / 3 · FROM PAID TEAM

Use a regressor for regression

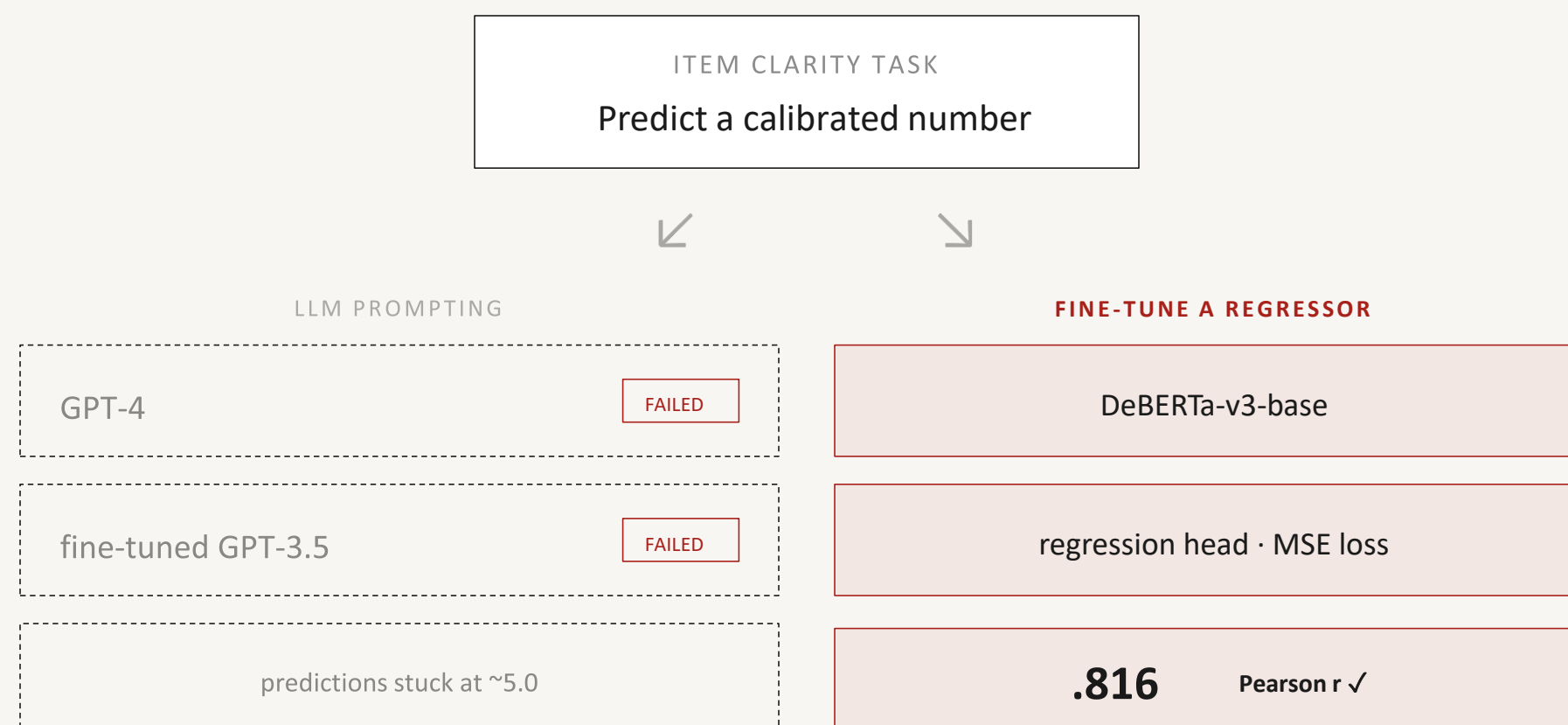
Use this when the target is a calibrated continuous value.
value.

“Tried GPT4 (failed). Tried fine-tuned GPT3.5 (failed).”

PAID TEAM · COMPETITION DECK

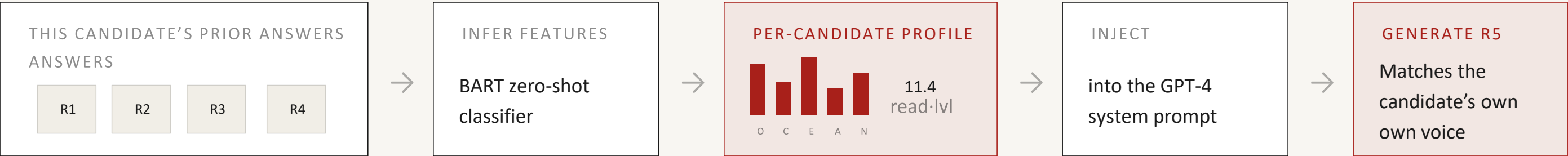
THE LESSON

LLMs can't do calibrated regression. The right tool is a small fine-small fine-tuned BERT-family regressor — ~140M params, 15
15 minutes on one GPU.



Condition on the input, do not pool

Use this when the metric is similarity to a per-instance reference.



WHY IT WORKS

The metric rewards style preservation, not content quality. Hungry Llama inferred inferred per-candidate features and conditioned generation. **PAID pooled across pooled across candidates and finished last.**

Hungry Llama

ForsMarsh

What this repo does — and what the scores mean

Reference implementations of all four tasks, run end-to-end on self-generated **synthetic inputs** . The official inputs were portal-gated and never archived, so the pipeline is archived, so the pipeline is validated — not benchmarked.

WHAT RUNS

4 / 4 TASKS

- ✓ All four task adapters, end-to-end on synthetic inputs
- ✓ OpenAI dispatch with retries, caching, structured outputs
- ✓ Scoring harness reproduces PAID’s .666 from published per-task scores

WHAT’S MISSING

THE REAL INPUTS

- ✗ Official input CSVs — portal-gated, never publicly archived
- ✗ Labels are public; the matching inputs are not
- ✗ So: a real, comparable benchmark score is out of reach



Three numbers, three meanings

What the winner scored, what this reimplementation would likely score on the real benchmark, and what it actually scored on scored on synthetic inputs.

● MEASURED

.666

PAID Team • 1st place
official competition test set

EMPATHY .580	INTERVIEW .440
CLARITY .816	FAIRNESS .828

● PROJECTED

~.65

This repo • real-data estimate
official inputs unavailable to run

EMPATHY ~.58	INTERVIEW ~.48
CLARITY ~.70	FAIRNESS ~.82

● SYNTHETIC

.817

This repo • synthetic inputs
⚠ not comparable to winners

EMPATHY 1.000	INTERVIEW .309
CLARITY .959	FAIRNESS 1.000

Projected = published winner scores as a ceiling, +1–2 pts for GPT-4o & structured outputs, –the LLM-only clarity ceiling ($r \approx .70$ vs. DeBERTa’s .816); priors, not measurements. **Synthetic .817** proves the pipeline runs end-to-end proves the pipeline runs end-to-end but is inflated by trivially-separable templates (dev split .814) — never rank it against the official numbers.

Projected position on the 2024 leaderboard

RANK	TEAM	INSTITUTION	COMPOSITE
1	PAID Team		.666
2	Akben & AAron		.652
~3	This repo PROJECTED	2024-evaluate-llms...	~.65
3	Hungry Llama	ForsMarsh	.643
4	Wonderlic ML		.630

Top of the official 2024 leaderboard. Projected row is not a measured rank.

THE PROJECTED MIDPOINT

Sits between **Akben (.652)** and **Hungry Llama Llama (.643)** — a gap of just **.009**.

Real rank depends on which tasks the harness wins on the official test set.

→ see slide 14 for the failure modes that move it

What cost winners real points

$$0.25 \times (\text{empathy} + \text{interview} + \text{clarity} + \text{fairness})$$

−4 pts

Using an LLM for calibrated regression

PAID’s deck: “Tried GPT4 (failed).” Chat models cluster predictions predictions around 5.0 with too-small variance. Fix: a fine-tuned BERT-tuned BERT-family regressor.

clarity_r

−7 pts

Pooling responses across candidates

PAID pooled and finished last on interview (.440) while Hungry Llama Llama preserved per-candidate style and won (.512). The metric metric rewards style, not content.

interview_cos

−1−3 pts

Skipping self-consistency on noisy labels

Akben tied PAID on fairness and beat them on empathy with N=5 majority-vote. Cost is 5× API calls; the gain is reliability near the model’s ceiling.

empathy · fairness

−2 pts

Random few-shot when similarity is available

Winners used all training examples as few-shot; picking K nearest neighbours by cosine similarity helps when N is large. Caveat: it *hurts* on clarity.

empathy_acc

What four winners collectively showed

✓ Reasoning supervision is cheap and good

PAID's Step-1/Step-2 trick — explain the training labels first, then reuse them as conversation-style few-shot — is one cached pass and reliably adds accuracy.

✓ Pairwise comparison beats absolute judgment

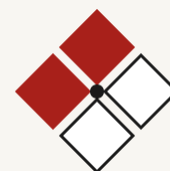
Akben's Elo trick is the genuinely creative move of the competition. It generalises to any noisy-label rating task where raters disagree.

✓ Self-consistency is a free 1–3 points

N=5 majority-vote at non-zero temperature. Akben's tied-1st fairness and 1st-place empathy scores both came from exactly this.

✓ Know when not to use an LLM

PAID abandoned GPT on clarity and fine-tuned DeBERTa instead, winning instead, winning the task by 4 Pearson points. LLMs don't do calibrated regression.



Four winners. Four approaches. No single right answer.